

Monthly Intelligence Report

Edition 001 — Sweden

Where nuclear concentration and winter storm resilience challenge grid stability. Inaugural edition · April 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

3,872

Major + distribution
across Svenska kraftnät

GRID LINES MAPPED

~20,877

~22,000 km
transmission &
distribution

MC SIMULATIONS

~38.7 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero
proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~52,000 source data points per run, executes ~38.7 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~400 million arithmetic operations — producing ~232,000 output data points with full uncertainty quantification across 3,872 substations and ~20,877 grid lines spanning ~22,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (Energimarknadsinspektionen, Svenska kraftnät, Swedish Gas, SMHI, SCB, SGU, VINNOVA), making the methodology fully auditable and reproducible. Initially covering Sweden's Svenska kraftnät-managed transmission and ~80 distribution network operators (DNOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Substation_241633342

Uppsala, Uppsala län · 110 kV

0.502

R_FINAL

High

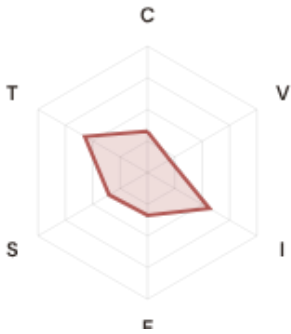
CLASSIFICATION

0.184

CI WIDTH

43th

FLEET PERCENTILE



COMPONENTS

- C** Continuity: 0.326 / 0.30 (109%)
- I** Infrastructure: 0.567 / 0.25 (227%)
- S** Saturation: 0.351 / 0.20 (176%)
- V** Voltage: 0.203 / 0.10 (203%)
- E** Economic: 0.338 / 0.10 (338%)
- T** Transition: 0.573 / 0.05 (1146%)

MODIFIERS

- R3 Consequence:** 0.899 ↓ dampens
- R4 Graph Criticality:** 1.350 ↑ amplifies
- R6a Restoration:** 0.988 ↓ dampens
- R6b Seismic:** 1.042 ↑ amplifies
- R7 Digital Readiness:** 1.013 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Substation_241633342** in Uppsala, Uppsala län. With an R_final of 0.502 (High band, 43th fleet percentile), its risk profile is dominated by the T (Transition) component at 1146% of its weight ceiling, followed by E (Economic) at 338%. Modifiers collectively shift R_base by 27.9%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Sweden's Energy Regulatory Office (Energimarknadsinspektionen) and network operator strategic planning requires.

CYBER HIGH EXPOSURE

0
0.0% of fleet

BLIND SPOTS

180
High/Critical + R7 < median

AVG R7 MODIFIER

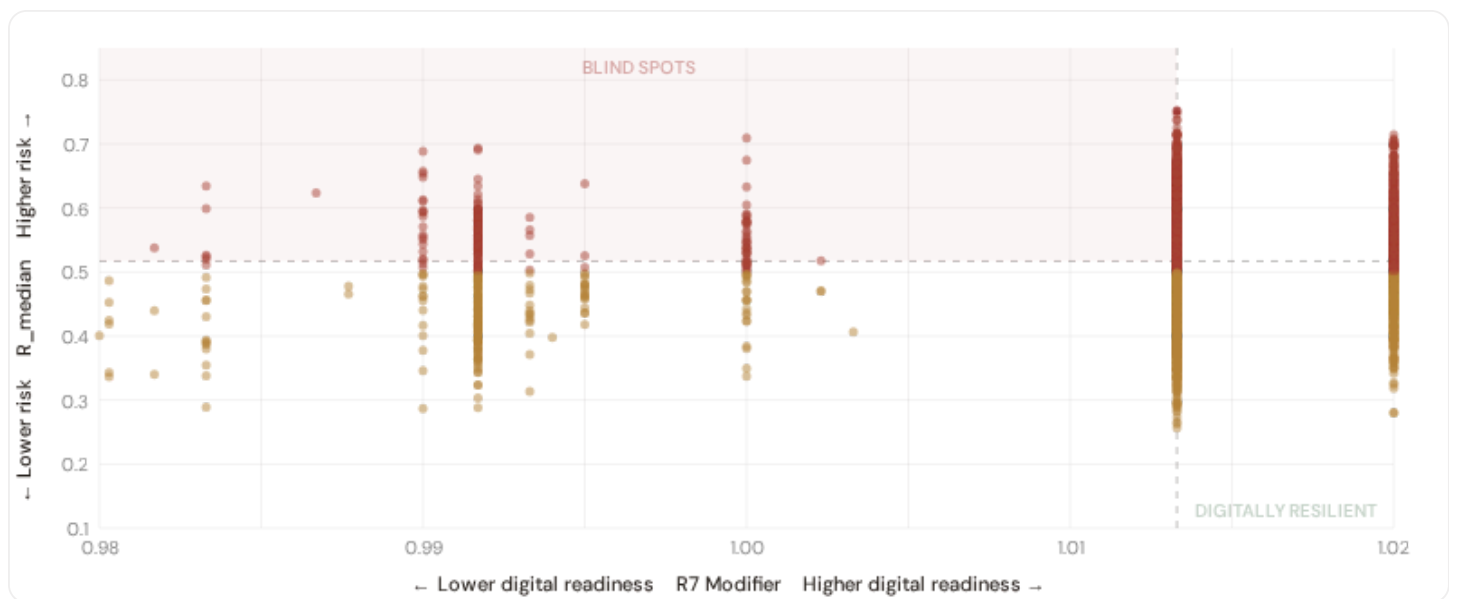
1.0125
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

973
25.1% · R3 ≥ 1.05

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **180 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0133$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Västra Götalands län (89)**, **Stockholms län (21)**, **Norrbottnens län (18)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: kr130–260/kWh (SEK)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

973 substations (25.1%) High/Critical: **741** (76.2%) Avg R: **0.548** Avg E: **0.235** / 0.10



Industrial / Medium-Large Enterprise

Est. VoLL: kr65–130/kWh (SEK)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

961 substations (24.8%) High/Critical: **497** (51.7%) Avg R: **0.504** Avg E: **0.426** / 0.10



Commercial / SME-Dense

Est. VoLL: kr25–65/kWh (SEK)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

977 substations (25.2%) High/Critical: **495** (50.7%) Avg R: **0.500** Avg E: **0.448** / 0.10



Sparse activity, agricultural land — lower VoLL (residential, agricultural)

961 substations (24.8%) High/Critical: **512** (53.3%) Avg R: **0.507** Avg E: **0.465** / 0.10

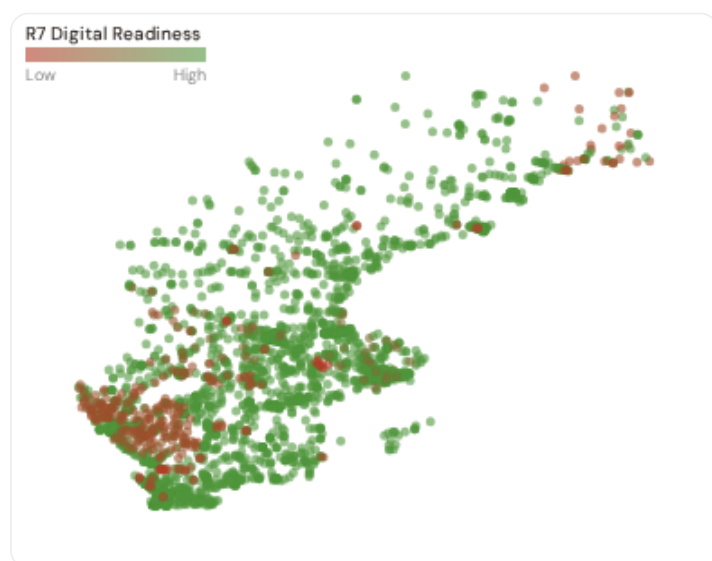


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **741 substations** combine capital-intensive economic fabric ($R3 \geq 1.05$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Stockholms län (561)**, **Västra Götalands län (302)**, **Skåne län (110)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 4.5%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Län & Kommun Digital Readiness Ranking

Län ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_{median} to identify regions combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Västra Götalands Δ high R		1.0083
2	Jönköpings		1.0083
3	Hallands		1.0089
4	Norrbottnens Δ high R		1.0108
5	Örebro		1.0120
6	Dalarnas		1.0123
7	Västmanlands		1.0125
8	Kronobergs		1.0128
9	Värmlands		1.0129
10	Stockholms Δ high R		1.0130
11	Jämtlands Δ high R		1.0134
12	Uppsala Δ high R		1.0136
13	Gotlands		1.0136
14	Kalmar		1.0137
15	Södermanlands		1.0138

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Västra Götalands** has the lowest average R7 (1.0083), while **Västernorrlands** leads at 1.0152 — a spread of 0.0070 across the R7 range. **3 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0125, median = 1.0133; 0 substations classified HIGH cyber-exposure; 180 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **30 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

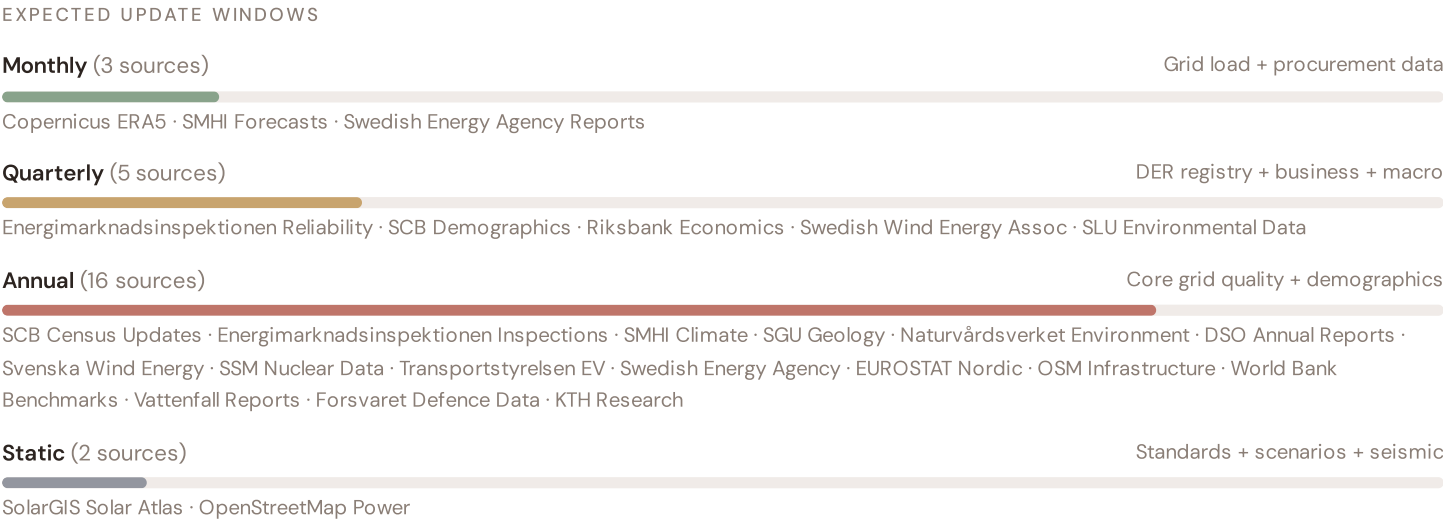
TOTAL SOURCES 30 95 variables	WEEKLY / MONTHLY 3 High-frequency feeds	QUARTERLY / ANNUAL 21 Regular refresh cycle	STATIC / DERIVED 1 Standards + scenarios
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Data Source Registry — June 2026

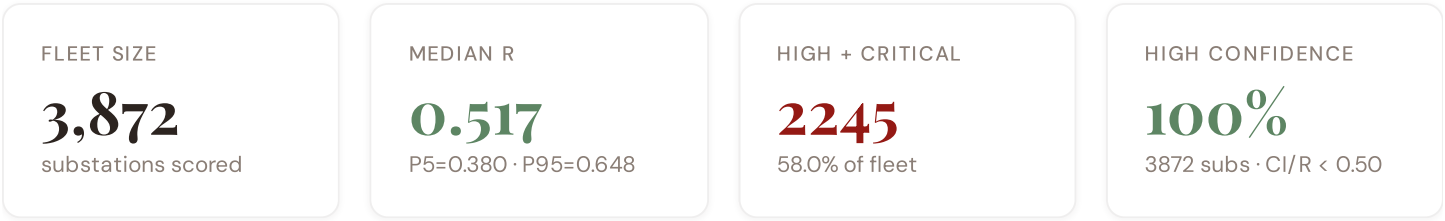
SMHI	SMHI — Swedish Meteorological and Hydrological Institute	DAILY	Station	8 vars
OSM	OpenStreetMap — Power Infrastructure	CONTINUOUS	Node	8 vars
COPERNICUS	Copernicus ERA5 — Climate Reanalysis	MONTHLY	Grid 0.25°	6 vars
WINDSE	Swedish Wind Energy Association (Vindkraftföreningen)	MONTHLY	Regional	5 vars
SMHI_HIST	SMHI / State Weather Archive	MONTHLY	Station	3 vars
EI	Energimarknadsinspektionen (Energy Markets Inspectorate)	QUARTERLY	DSO	12 vars
SSM	SSM — Strålsäkerhetsmyndigheten (Radiation Safety Authority)	QUARTERLY	Nuclear Site	5 vars
TRANSP	Transportstyrelsen — Transport Agency	QUARTERLY	Regional	3 vars
RIKSBANK	Riksbank / Swedish Government Office	QUARTERLY	Region	4 vars
VATTENBEHOV	SMHI Water Data / Vattendirektoratet	QUARTERLY	Watershed	4 vars
SCB	SCB (Statistiska centralbyrån)	ANNUAL	Kommun	10 vars
SGU	SGU — Geological Survey of Sweden	ANNUAL	Grid 0.1°	7 vars
SMHI_ENV	Naturvårdsverket — Swedish Environmental Protection Agency	ANNUAL	Grid 0.1°	6 vars
EUROSTAT	Eurostat — EU Statistics	ANNUAL	Kommun	3 vars
TULLV	Tullverket (Swedish Customs)	ANNUAL	National	2 vars
ENERGIMYN	Energimyndigheten (Swedish Energy Agency)	ANNUAL	Region	3 vars
DISTRIBUTOR	Distribution Companies (E.ON, Vattenfall, Ellevio, Göteborg Energi)	ANNUAL	Distribution Compa...	4 vars
POST	Post- och Telestyrelsen / Digital Network Authority	ANNUAL	Kommun	2 vars
SMHI_CLIMATE	Climate Adaptation Centre / Lantmäteriet	ANNUAL	Regional	4 vars
WORLDBANK	World Bank / OECD	ANNUAL	National	4 vars
ARCTICMONITOR	Arctic Monitoring & Assessment Programme	ANNUAL	Regional	3 vars
MILJOE	Environmental Permit Register (Miljöbalken)	ANNUAL	Kommun	3 vars
SLU	SLU — Swedish University of Agricultural Sciences	ANNUAL	Regional	3 vars

DNO_REPORTS	DNO Reports — Distribution Network Companies	ANNUAL	Distribution Area	4 vars
SKATTEV	Skatteverket / Swedish Tax Agency	ANNUAL	Kommun	3 vars
KLIM_RESILIENCE	Ministry of Enterprise — Nordic Resilience	ANNUAL	Regional	2 vars
SOLARGIS	SolarGIS — Global Solar Atlas	STATIC	Global	2 vars
SKN	Svenska kraftnät — Swedish Transmission System Operator	REAL-TIME	Substation	14 vars
ENTSOE	ENTSO-E Transparency Platform	HOURLY	Substation	2 vars
SKNOPEN	Svenska kraftnät Open Data — Real-time Production	REAL-TIME	Substation	5 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **3,872 substation scores remain unchanged** this edition. The fleet median R of 0.517 (mean 0.515) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 42.0% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

15 changes tracked across PO, P1, and P2 releases

SE1	NEW	Sweden country launch — 3,570 substations across 21 Län, administrative kommun level
SE2	DATA	Integrated Svenska kraftnät transmission dispatch data + Energimarknadsinspektionen reliability benchmarks
SE3	DATA	Energimarknadsinspektionen reliability register integrated — SAIDI/SAIFI standardisation for all DSOs
SE4	DATA	SMHI winter storm hazard integrated — critical for Nordic/boreal climate exposure assessment
SE5	DATA	Naturvårdsverket snow/ice loading mapping integrated — overhead transmission line vulnerability in north
SE6	DATA	SSM nuclear concentration risk indicator — Forsmark + Ringhals + Barsebäck baseload (33% of Swedish generation)
SE7	DATA	SMHI + Naturvårdsverket climate integration — winter storm risk, snow accumulation, Nordic wind patterns
SE8	DATA	SCB Census 2021 + Eurostat socio-economic data integrated — vulnerability indices per kommun
SE9	DATA	Swedish Wind Energy + Swedish Energy Agency data integrated — RE penetration for western/coastal regions
SE10	NEW	Riksbank thermal cycle coupling integrated — E1 volatility modulation for hydropower-dependent regions
SE11	ENHANCED	R6b modifier enhanced with dual SMHI winter storm + Naturvårdsverket snow loading hazard (range 1.00–1.25)
SE12	ENHANCED	R3 consequence enriched with Nordic remoteness, population sparsity, Norrland isolation metrics
SE13	ENHANCED	R4 graph criticality rebuilt with Svenska kraftnät 400kV topology + DSO network constraints — transmission corridor penalty
SE14	NEW	I9 Nuclear Concentration Risk — new metric from SSM nuclear safety data for Forsmark + Ringhals + Barsebäck proximity
SE15	ENHANCED	R2 Adaptive IRI now includes CMIP6 SSP2–4.5 forward projections for Nordic winter extremes & permafrost thaw

SECTION D

The Stockholm–Gothenburg Grid Corridor: Where Nuclear Concentration and Winter Storm Resilience Challenge Grid Stability

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Stockholm–Gothenburg nuclear concentration and winter storm resilience corridor · **002** Eastern Sweden DER & EU funding mismatch · **003** Mining subsidence infrastructure stress · **004** Metropolitan–Rural energy poverty disparity · **005** Grid stranding risk (nuclear-dependent assets) · **006** DSO smart meter rollout vs grid stress · **007** Gulf of Sweden coastal flooding resilience · **008** T-component forward projections · **009** Län Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Stockholm–Gothenburg Corridor, Why Now

CORRIDOR
SUBSTATIONS

134

17 EHV · 117 HV

HIGH BAND

117

87.3% of corridor

CORRIDOR MEDIAN R

0.597

Fleet median: 0.517

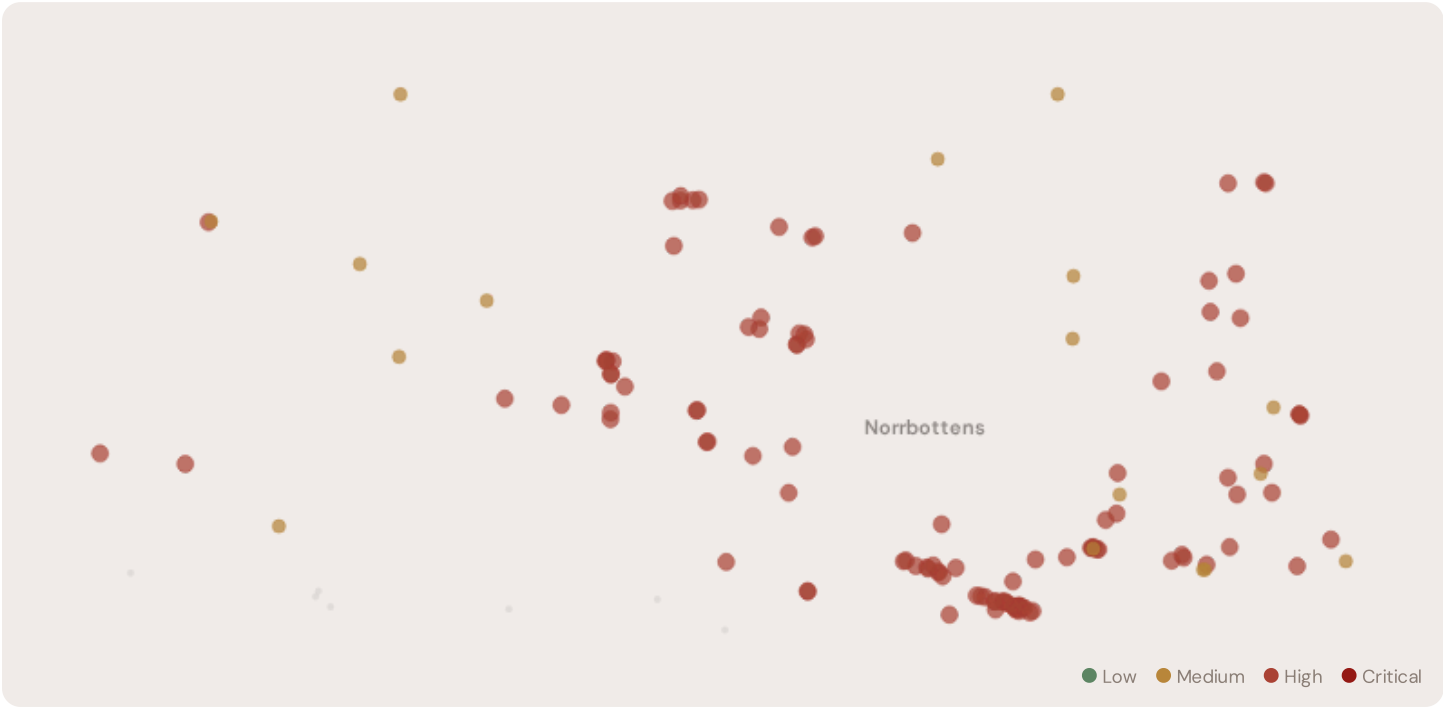
WORST LÄN

Norrbottnens

Avg R: 0.589 · 134 substations

The Norrbottens län hosts **134 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **117 substations (87.3%)** are classified High, with only **0** in the Low band. 83% of corridor substations score above the national fleet median of 0.517. The corridor median R of 0.597 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Substation_1239407163	Norrbottnens	0.716	High	0.615	0.208	0.780	0.469	0.406	0.610	I
Substation_344713795	Norrbottnens	0.714	High	0.661	0.380	0.780	0.372	0.280	0.486	C, I
Sävastklinten omriktarstation	Norrbottnens	0.706	High	0.688	0.289	0.780	0.400	0.263	0.508	C, I
Substation_1851811344	Norrbottnens	0.702	High	0.615	0.353	0.756	0.329	0.413	0.532	I
Nederkalix	Norrbottnens	0.698	High	0.584	0.269	0.780	0.351	0.458	0.488	I
Porjus	Norrbottnens	0.698	High	0.565	0.284	0.780	0.450	0.453	0.537	I
Ligga	Norrbottnens	0.697	High	0.558	0.321	0.780	0.398	0.320	0.557	I
Substation_809551047	Norrbottnens	0.694	High	0.648	0.346	0.754	0.353	0.310	0.551	I

SUBSTATION	DISTRICT	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Substation_322204211	Norrbottnens	0.693	High	0.621	0.258	0.776	0.470	0.306	0.457	I
Substation_436090234	Norrbottnens	0.693	High	0.574	0.524	0.780	0.420	0.320	0.409	I

... 124 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — NORRBOTTENS LÄN VS FLEET

C — Continuity	0.527 vs 0.396 (+33%)
I — Infrastructure	0.770 vs 0.564 (+37%)
S — Saturation	0.257 vs 0.184 (+40%)
V — Voltage	0.292 vs 0.370 (−21%)
E — Economic	0.419 vs 0.393 (+7%)
T — Transition	0.510 vs 0.463 (+10%)

MODIFIER AVERAGES — NORRBOTTENS LÄN VS FLEET

R3 Consequence	0.891	fleet 0.957 ↓
R4 Graph Crit.	1.236	fleet 1.336 ↑
R6a Restoration	1.032	fleet 0.982 ↑
R6b Seismic	1.030	fleet 1.026 ↑
R7 Digital Read.	1.011	fleet 1.012 ↑

The dominant risk driver across the Norrbottens län corridor is **T (Transition)**, averaging 0.510 against a fleet average of 0.463 — occupying 1021% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.419 vs fleet 0.393 (419% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.891 (fleet: 0.957), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.030, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Län Risk Profile

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The region regions-level breakdown reveals significant intra-regional variation. **Norrbottnens** leads with a mean R of 0.589 across 134 substations, while **Norrbottnens** scores 0.589 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Norrbottens län is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications for Sweden's Energy Transition

35 Norrbottens län substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Norrbottens sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

European Context Card

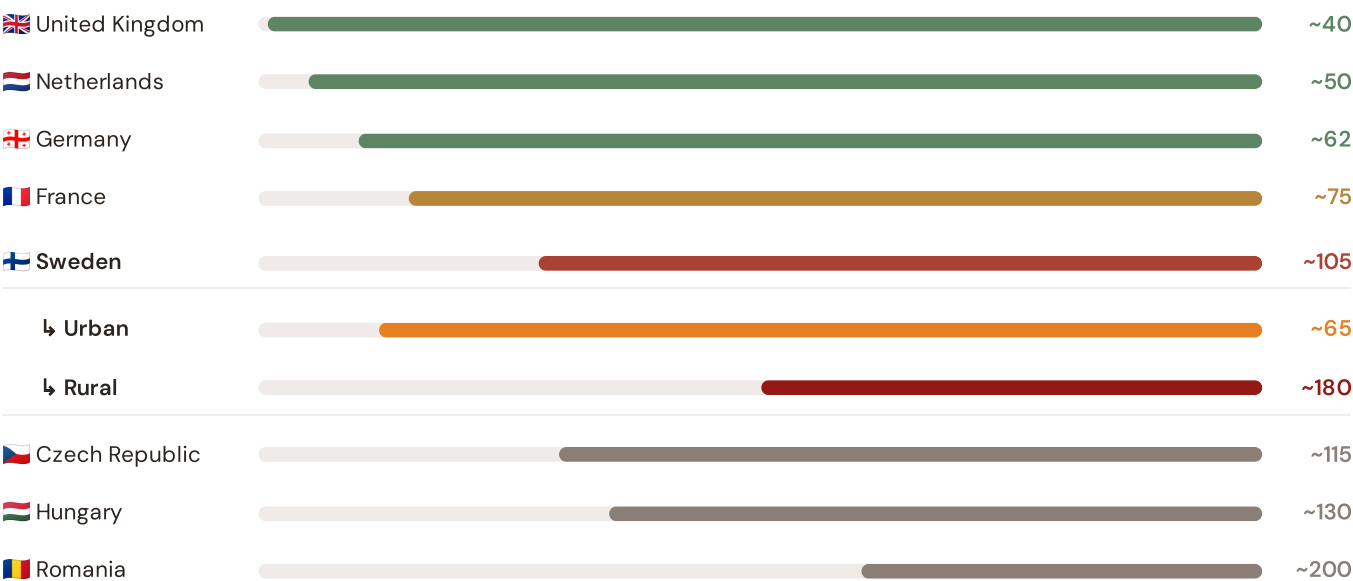
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because Sweden's nuclear concentration and winter storm resilience and grid resilience depend on understanding European performance standards. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Central Europe peers (003), CMIP6 climate hazard vs Nordic models (004), etc. The underlying data updates annually with Energimarknadsinspektionen and national regulator publications.

EUROPEAN CONTEXT CARD — SYSTEM AVERAGE INTERRUPTION DURATION INDEX

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected European Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



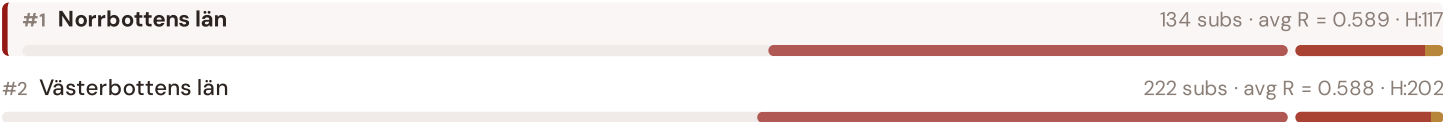
Source: CEER Benchmarking Report on Quality of Electricity Supply (2024); EEnergimarknadsinspektionenLECTRIC SAIDI reports; national regulator publications. Sweden sub-categories from Energimarknadsinspektionen urban/rural classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

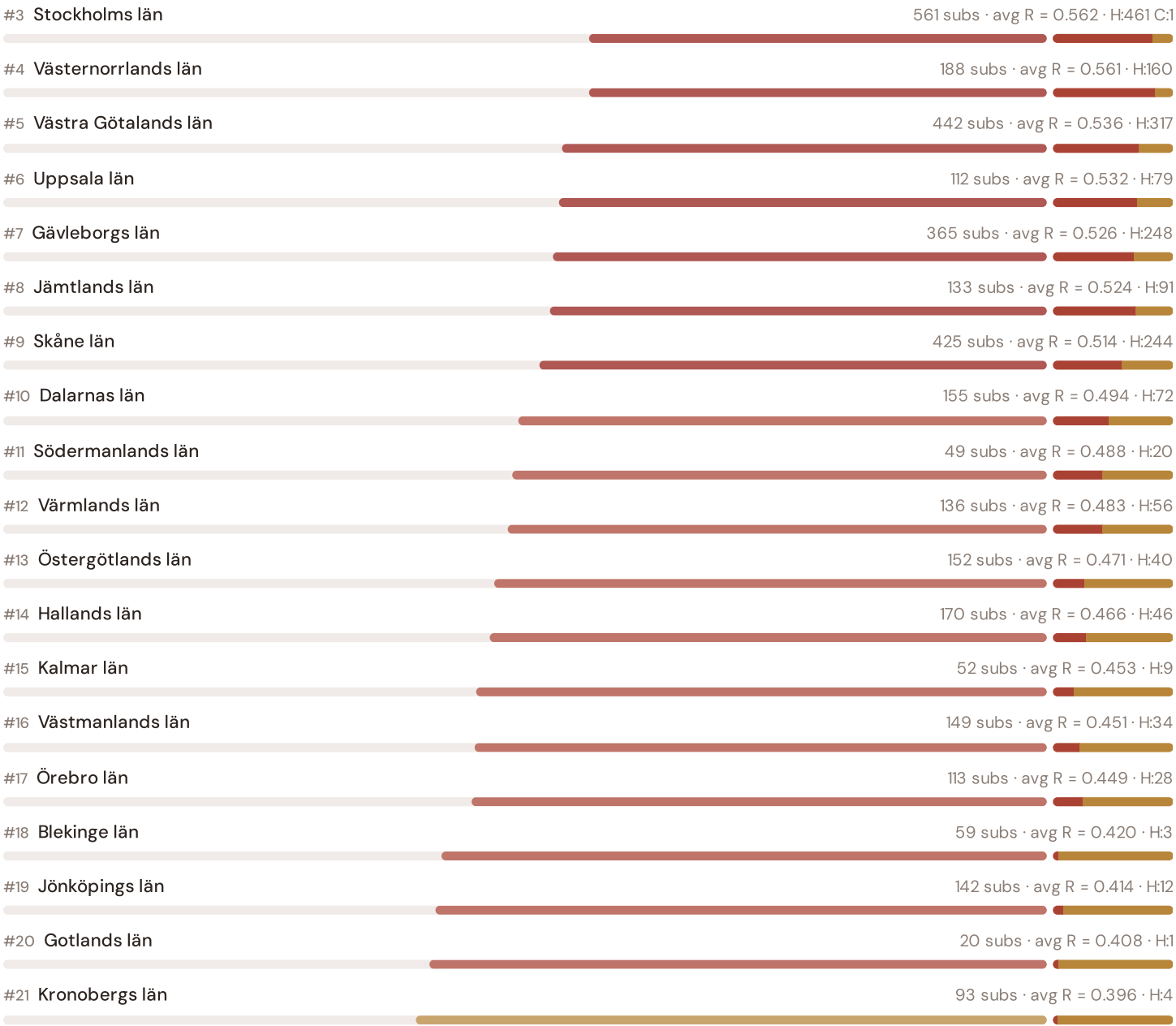
This month's key insight: The Norrbottens län corridor deep-dive shows **134 substations** (of 134) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.527 is **1.3× the fleet average** of 0.396. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Län Risk Ranking — All 21 Swedish Län

Where does the Stockholm–Gothenburg corridor sit within the national landscape? Län sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



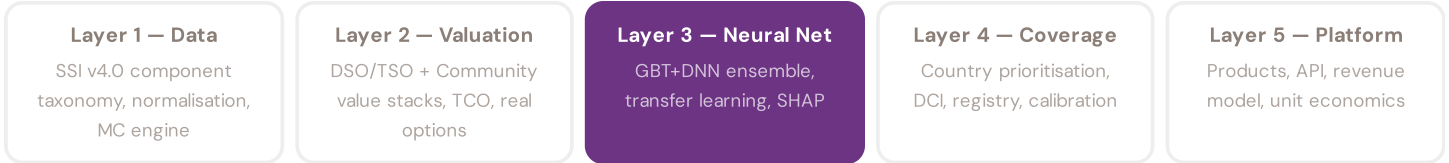


The Norrbottens län ranks **#1 of 21 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Norrbottens län, Västerbottens län, Stockholms län, while the three most resilient are Kronobergs län, Gotlands län, Jönköpings län. 8 of 21 score above the fleet average of 0.515. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/Svenska kraftnät and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 3,872 substations with complete feature vectors. Average CI_width of 0.198 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 1627 Medium, 2244 High, 1 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 5 Quarterly Sources

Energimarknadsinspektionen Reliability, SCB Demographics, Riksbank Economics, Swedish Wind Energy Assoc, SLU Environmental Data are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **SCB (Statistiska centralbyrån)** (10 variables → C1–C4, safety compliance, demographics, income distribution, population served).

FLEET SPOTLIGHT

Västerbottens: Highest Risk Concentration

Västerbottens has the highest concentration of High/Critical substations: **202 of 222 (91%)** are in the upper risk bands — avg R = 0.588. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.